

ASSESSMENT CO2- AT2

DEBUGGING

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Course Code: CSA0312

Course Name: Data Structures

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=539`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and a "DS PROGRAMMING" badge. The user profile "Bedudoori Siva Parvathi" with ID "192525194RC" is visible in the top right.

On the left, a sidebar indicates "1/3 answered". The main area is divided into three sections:

- Your Fixed Code:** A code editor containing the following C code:

```
1 #include <stdio.h>
2 int linearSearch(int arr[], int n, int key) {
3     for(int i = 0; i < n; i++) {
4         if(arr[i] == key)
5             return i;
6     }
7     return -1;
8 }
9 int main() {
10    int arr[] = {10, 20, 30, 40, 50};
11    int n = 5;
12    int key = 30;
13    int result = linearSearch(arr, n, key);
14    if(result != -1)
15        printf("Element found at index %d", result);
16    else
17        printf("Element not found");
18    return 0;
19 }
```
- Test Input:** A text area containing the input:

```
5
10 20 30 40 50
20
```
- Output:** A dark box displaying the result:

```
Element found at index 2
```

At the bottom of the code editor, there are three buttons: "Run" (with a play icon), "Save" (with a save icon), and "Submit Fix" (with a star icon).

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QUESTIONS

Q1
1. Find the bugs, include the missing lines in the following snippet.
1 mark

Q2
2. Buggy Code in an array
1 mark

Q3
Identify an errors in the following snippet
1 mark

2/3 answered

Q2 2. Buggy Code in an array 1 mark

```
#include <stdio.h>

int main() {
    int arr[5] = {10,20,30,40,50};
    for(int i=0;i<=5;i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Expected output:
10 20 30 40 50

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int main() {
3     int arr[5] = {10, 20, 30, 40, 50};
4     for (int i = 0; i < 5; i++)
5         printf("%d ", arr[i]);
6     return 0;
7 }
```

Test Input

stdin input (optional)

Output

10 20 30 40 50

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```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int key){
3     int low = 0, high = n - 1;
4     while (low <= high){
5         int mid = (low + high) / 2;
6         if (arr[mid] == key)
7             return mid;
8         else if (arr[mid] < key)
9             low = mid + 1;
10        else
11            high = mid - 1;
12    }
13    return -1;
14 }
15 int main(){
16     int n, key, i, pos;
17     printf("Enter size: ");
18     scanf("%d", &n);
19     int arr[n];
20     printf("Enter sorted elements: ");
21     for(i = 0; i < n; i++)
22         scanf("%d", &arr[i]);
23     printf("Enter key: ");
24     scanf("%d", &key);
25     pos = binarySearch(arr, n, key);
26     if(pos != -1)
27         printf("Element found at index %d", pos);
28     else
29         printf("Element not found");
30     return 0;
31 }
```

5
10 20 30 40 50
30

Output

Enter size: Enter sorted
elements: Enter key: Element
found at index 2